ECO 3121 2024 – 2025 Fall Assignment 3

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ECO3121: Introductory Econometrics

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Question 1

1. **Stata Codes:**

```
* Preparation
keep if year == 2001

gen yield = hx95_nh125 / hx95_nh112
gen fertilizer = nh6_3 / hx95_nh112

* 1. Generate a binary variable regarding household migration decision.
gen migrate = h95_nh269 > 0

* Generate the natural log of yield and fertilizer application intensity
gen log_yield = log(yield + 1)
gen log_fertilizer = log(fertilizer + 1)
```

Required Output:

```
. * Preparation
. keep if year == 2001
(38,832 observations deleted)

.
. gen yield = hx95_nh125 / hx95_nh112
(1,268 missing values generated)

. gen fertilizer = nh6_3 / hx95_nh112
(1,268 missing values generated)

.
. * 1. Generate a binary variable regarding household migration decision.
. gen migrate = h95_nh269 > 0

.
. * Generate the natural log of yield and fertilizer application intensity
. gen log_yield = log(yield + 1)
(1,268 missing values generated)

. gen log_fertilizer = log(fertilizer + 1)
(1,268 missing values generated)
```

2. **Stata Codes:**

```
reg migrate log_yield
```

Required Output:

```
. * 2. Run a linear probability model of household migration decision in question (1) on the natural log of yield.
. reg migrate log_yield
```

Source	SS	df	MS	Number of obs	=	4,200
				F(1, 4198)	=	14.88
Model	3.68868145	1	3.68868145	Prob > F	=	0.0001
Residual	1040.44251	4,198	.247842427	R-squared	=	0.0035
				Adj R-squared	=	0.0033
Total	1044.13119	4,199	.24866187	Root MSE	=	.49784

migrate	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
log_yield	.0495398	.0128412	3.86	0.000	.0243642	.0747155
_cons	.1773844	.0743338	2.39	0.017	.0316509	.3231179

Estimators:

Intercept (_cons): 0. 1773844 (Std. Err.: 0.743338)

log_yield: 0.0495398 (Std. Err.: 0.0128412)

Interpretation:

Holding other factors fixed, the migrate increases by (0.495398 / 100) for every 1% increase in yield.

Comment on its statistical significance:

Because the p-value for variable log_yield is smaller than 0.000, indicates that the variable log_yield is significantly different from 0 at the 1% significance level.

3. **Three plausible arguments:**

Omitted Variable Bias: Variables like household wealth might affect both yield and migration, causing Upward bias.

Measurement Error: Inaccurate yield measurement may bias the estimate Downward.

Simultaneity: Migration might affect yield, causing endogeneity to be a potential bias. More specifically, we can use mathematics proof to find out.

$$migrate_i = \beta_0 + \beta_1 \cdot yield_i + \varepsilon_i$$

$$yield_i = \gamma_0 + \gamma_1 \cdot migrate_i + u_i$$

$$\widehat{\beta}_1 = \beta_1 + \rho \cdot \frac{\sigma_\varepsilon}{\sigma_{yield}} \qquad where\ \rho = Corr(\varepsilon, yield)$$

If ρ > 0, the bias is Upward.

If ρ < 0, the bias is Downward.

4. **Stata Codes:**

```
* 4. Using stata command merge
merge m:1 vl_id using village_rainfall.dta

* You decide to use the natural log of average rainfall in 2001 (log(av rain)) as the IV for the natural log of yield
gen log_av_rain = log(av_rain)

* Make sure the 2SLS is operate on the same, valid dataset.
drop if missing(log_yield, log_av_rain, migrate)

* Write down the First stage regression model
reg log_yield log_av_rain

* Produce the log_yield_hat
predict log_yield_hat

* Check whether the all the data are valid.
summ log_yield_hat
list if missing(log_yield_hat)
```

Required Output:

```
. * 4. Using stata command merge
. merge m:1 vl_id using village_rainfall.dta
```

Result	Number of obs	
Not matched	373	
from master	0	(_merge==1)
from using	373	(_merge==2)
Matched	5,468	(_merge==3)

```
.
. * You decide to use the natural log of average rainfall in 2001 (log(av rain)) as the IV for the natural log of yield
. gen log_av_rain = log(av_rain)
(448 missing values generated)

.
. * Make sure the 2SLS is operate on the same, valid dataset.
. drop if missing(log_yield, log_av_rain, migrate)
(1,838 observations deleted)

.
. * Write down the First stage regression model
. reg log_yield log_av_rain
```

Source	SS	df	MS		
Model	182.285667	1	182.285667	Number of obs	= 4,003
				F(1, 4001)	= 560.65
Residual	1300.86357	4,001	.325134609	Prob > F	= 0.0000
				R-squared	= 0.1229
				Adj R-squared	= 0.1227
Total	1483.14924	4,002	.370602008	Root MSE	= .57021

log_yield	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
log_av_rain	.4644991	.0196173	23.68	0.000	.4260382	.50296
_cons	2.526537	.1365663	18.50	0.000	2.258791	2.794284

```
. * Produce the log_yield_hat
. predict log_yield_hat
(option xb assumed; fitted values)

.
. summ log_yield_hat
```

Variable	Obs	Mean	Std. dev.	Min	Max
log_yield_~t	4,003	5.753103	.2134213	5.320964	6.0853

```
. list if missing(log_yield_hat)
```

The specification of model is:

$$log_yield_i = \delta_0 + \delta_1 log_av_rain_i + \epsilon_i$$

Based on the data, the regression result of the model is:

$$\widehat{log_yield}_i = \underset{(0.1365663)}{2.526537} + \underset{(0.0196173)}{0.4644991}\, log_av_rain_i$$

Estimators:

Intercept (_cons): 2.526537 (Std. Err.: 0.1365663)

log_av_rain: 0.4644991 (Std. Err.: 0.0196173)

Interpret the result:

Holding other factors fixed, the yield increases by 0.4644991% for every 1% increase in av_rain.

Comment on its statistical significance:

Because the p-value for variable log_av_rain is smaller than 0.000, indicates that the variable log_av_rain is significantly different from 0 at the 1% significance level.

5. **Stata Codes:**

```
* 5. Write down the second stage regression modela
reg migrate log_yield_hat

* Compared to the first and second step of the 2SLS
count if !missing(log_yield, log_av_rain, migrate)
count if !missing(log_yield_hat, migrate)
```

Required Output:

```
. * 5. Write down the second stage regression modela
. reg migrate log_yield_hat
```

Source	SS	df	MS	Number of obs	=	4,003
				F(1, 4001)	=	140.16
Model	33.6398682	1	33.6398682	Prob > F	=	0.0000
Residual	960.267701	4,001	.240006924	R-squared	=	0.0338
				Adj R-squared	=	0.0336
Total	993.907569	4,002	.248352716	Root MSE	=	.48991

migrate	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
log_yield_hat	.4295868	.0362857	11.84	0.000	.3584465	.500727
_cons	-2.012801	.208899	-9.64	0.000	-2.422359	-1.603243

```
.
. * Compared to the first and second step of the 2SLS
. count if !missing(log_yield, log_av_rain, migrate)
  4,003

. count if !missing(log_yield_hat, migrate)
  4,003
```

The specification of model is:

$$migrate = \gamma_0 + \gamma_1 \widehat{log_yield}_i + \rho_i$$

Based on the data, the regression result of the model is:

$$\widehat{migrate}_i = \underset{(0.208899)}{-2.012801} + \underset{(0.0362857)}{0.4295868}\ \widehat{log_yield}_i$$

Coefficients of Estimators:

Intercept (_cons): -2.012801 (Std. Err.: 0.208899)

$\widehat{log_yield}_i$: 0.4295868 (Std. Err.: 0. 0362857)

Interpret the result:

Holding other factors fixed, migrate increases by 0.4295868% for every 1% increase in $\widehat{log_yield}_i$.

Comment on its statistical significance:

Because the p-value for variable $\widehat{log_yield}_i$ is smaller than 0.000, indicates that the variable $\widehat{log_yield}_i$ is significantly different from 0 at the 1% significance level.

6. **(a) Stata Codes:**

```
* 6. You that you can use ivregress 2sls command directly to replicate the results in question (5).
ivregress 2sls migrate (log_yield = log_av_rain)
```

Required Output:

```
. * 6. You that you can use ivregress 2sls command directly to replicate the results in question (5).
. ivregress 2sls migrate (log_yield = log_av_rain)

Instrumental variables 2SLS regression          Number of obs   =      4,003
                                                Wald chi2(1)    =     111.36
                                                Prob > chi2     =     0.0000
                                                R-squared       =          .
                                                Root MSE        =     .54961

------------------------------------------------------------------------------
     migrate | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
  log_yield |   .4295868   .0407081    10.55   0.000     .3498003    .5093732
      _cons |  -2.012801    .234359    -8.59   0.000    -2.472136   -1.553466
------------------------------------------------------------------------------
Endogenous: log_yield
Exogenous:  log_av_rain
```

Coefficients of Estimators:

Questions (5) Linear Regression Model:

Intercept (_cons): -2.012801 (Std. Err.: 0.208899)

$\widehat{log_yield}_i$: 0.4295868 (Std. Err.: 0. 0362857)

2SLS:

Intercept (_cons): -2.012801 (Std. Err.: 0.234359)

$\widehat{log_yield}_i$: 0.4295868 (Std. Err.: 0.0407081)

Conclusion:

By using two different ways, we get the same results with no difference, but the standard error is relatively larger than we got in (5). This is because in manual methods, the standard error of log_yield does not be considered the variability introduced in the first stage regression.

(b):

Comparing the different estimators by two different ways:

Linear Regression Model: 0.0495398

2SLS: 0.4295868

We can find out: If we ignore the endogeneity issue in the independent variable, the estimator will be biased, and the direction is Upward as we expected in the Question (3).

Question 2

1. **β_0**: Positive. More fertilizer should increase yield.

   β_1: Positive. Better yield may encourage more fertilizer use.

2. Simultaneity Bias: The two equations influence each other to a certain degree, leading to biased OLS estimates.

   We can use the mathematics method to prove the biased of the estimators:

$$Crop = \alpha_0 + \beta_0 * Fertilizer + u$$

$$Fertilizer = \alpha_1 + \beta_1 \times Crop + v$$

$$Crop = \alpha_0 + \beta_0 \times (\alpha_1 + \beta_1 \times Crop + v) + u$$

$$Crop = \frac{\alpha_0 + \beta_0 * \alpha_1 + \beta_0 * v + u}{1 - \beta_0 * \beta_1}$$

$$\hat{\beta}_0 = \beta_0 + \frac{Cov(Fertilizer, u)}{Var(Fertilizer)}$$

$$E[Fertilizer \mid u] \neq 0$$

$$Cov(Fertilizer, u) > 0$$

$$OLS\ of \hat{\beta}_0\ will\ larger\ than\ \beta_0, which\ means\ Upward.$$

Based on the same calculation, we can also get the expression of the estimator $\hat{\beta}_1$:

$$\hat{\beta}_1 = \beta_1 + \frac{Cov(Crop, v)}{Var(Crop)}$$

$$E[Crop \mid v] \neq 0$$

$$Cov(Crop, v) > 0$$

$$OLS\ of \hat{\beta}_1\ will\ larger\ than\ \beta_1, which\ means\ Upward.$$

3. For a proper instrumental variable Z, it should satisfy two conditions:

$$Cov(Z, X_i) \neq 0$$

$$Cov(Z, u) = 0$$

That's how we can find out the instrumental variable for each regression:

For the Variable Fertilizer, we choose the iv: Budget, which satisfies:

$$Cov(Budget, Fertilizer) \neq 0$$

$$Cov(Budget, u) = 0$$

For the Variable Crop, we choose the iv: Sunshine, which satisfies:

$$Cov(Sunshine, Crop) \neq 0$$

$$Cov(Sunshine, v) = 0$$

Appendix:

STATA do.file

```
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* 2. Run a linear probability model of household migration decision in question (1) on the natural log of yield.
reg migrate log_yield

* 3.
* Omitted Variable Bias: Variables like household wealth might affect both yield and migration, causing upward bias.
* Measurement Error: Inaccurate yield measurement may bias the estimate downward.
* Simultaneity: Migration might affect yield, causing endogeneity and potential bias.

* 4. Using stata command merge
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